MATHEMATICAL FRAMEWORK

for the

EU Food Fraud Vulnerability Intelligence System

Quantitative Models for Commodity Dependency,
Origin–Attention Country Trade Relationships,
and Hazard–Trade Corridor Integration

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1. Notation and Definitions

All models in this framework use the following standard notation. Every variable is grounded in observable, collectible data from the operational sources: RASFF Window, UN Comtrade, FAOSTAT, and Eurostat.

1.1 Index Sets

Index Sets

$$
\begin{aligned}
c &\in \mathcal{C} && \text{A commodity, identified by HS 6-digit code} \\
i &\in \mathcal{N} && \text{A destination (importing) country — the attention/follow-up country} \\
j &\in \mathcal{O} && \text{An origin (exporting) country} \\
t &\in \mathcal{T} && \text{A time period (year or year-month)} \\
h &\in \mathcal{H} && \text{A hazard type}
\end{aligned}
$$

where:

$$\mathcal{H} = \{\text{biological, chem_pesticides, chem_heavy_metals, chem_mycotoxins, chem_other, regulatory}\}$$

1.2 Primary Data Variables

1.2.1 From UN Comtrade

Trade Variables

$$M(c,i,j,t) = \text{Import quantity (kg) of commodity } c \text{ by country } i \text{ from origin } j \text{ in period } t$$

$$V(c,i,j,t) = \text{Import value (USD) of commodity } c \text{ by country } i \text{ from origin } j \text{ in period } t$$

$$M(c,i,\bullet,t) = \sum_{j \in \mathcal{O}} M(c,i,j,t) \quad \text{Total imports from all origins}$$

$$X(c,j,\bullet,t) = \sum_{i \in \mathcal{N}} M(c,i,j,t) \quad \text{Total exports from origin } j \text{ to all destinations}$$

1.2.2 From FAOSTAT / Eurostat

Production & Consumption Variables

$$P(c,i,t) = \text{Domestic production quantity (kg) of commodity } c \text{ in country } i$$

$$D(c,i,t) = \text{Domestic supply for food use (apparent consumption) of commodity } c \text{ in country } i$$

$$\text{Pop}(i,t) = \text{Population of country } i \text{ in period } t$$

1.2.3 From RASFF Window

Hazard Signal Variables

$$
\begin{aligned}
R(c,i,j,t) &= \text{Count of RASFF notifications for commodity } c \text{ involving origin } j \\
&\qquad \text{where country } i \text{ is notifier, attention, or follow-up country, in period } t \\
R_h(c,i,j,t) &= \text{Count of RASFF notifications filtered to hazard type } h \\
S(r) &= \text{Severity weight assigned to notification } r \text{ (defined in §4.2)}
\end{aligned}
$$

2. Commodity Dependency Models

This section defines the mathematical models for quantifying how dependent a destination country (attention/alert country) is on a specific origin country for a specific commodity. This is the structural vulnerability layer—it captures the conditions that make fraud economically viable and difficult to resist.

2.1 Domestic Supply Balance

The foundation of all dependency analysis is the commodity supply balance equation.

Definition 2.1 (Domestic Supply)**.** For commodity c in country i at time t:

$$\boxed{\mathrm{DS}(c,i,t) \;=\; P(c,i,t) \;+\; M(c,i,\bullet,t) \;-\; X(c,i,\bullet,t) \;+\; \Delta S(c,i,t)} \tag{1}$$

where $\Delta S(c,i,t)$ is the change in stocks (draw-down positive, build-up negative).

When stock change data is unavailable (common at commodity level), set $\Delta S = 0$ to obtain **apparent domestic supply**:

$$\mathrm{DS}'(c,i,t) \;=\; P(c,i,t) + M(c,i,\bullet,t) - X(c,i,\bullet,t) \tag{2}$$

This is the standard FAO simplification used in Food Balance Sheets. Where FAOSTAT provides pre-computed balance data, use it directly. Where it does not, compute from Comtrade (imports, exports) and FAOSTAT/Eurostat (production).

2.2 Import Dependency Ratio (IDR)

Definition 2.2 (Import Dependency Ratio)**.** The share of country i's total domestic supply of commodity c that comes from imports:

$$\boxed{\mathrm{IDR}(c,i,t) \;=\; \frac{M(c,i,\bullet,t)}{\mathrm{DS}'(c,i,t)} \;=\; \frac{M(c,i,\bullet,t)}{P(c,i,t) + M(c,i,\bullet,t) - X(c,i,\bullet,t)}} \tag{3}$$

Property 2.1 (IDR Interpretation)**.**

- $\mathrm{IDR} = 0$: Fully self-sufficient. All supply is domestic production.
- $\mathrm{IDR} = 1$: Fully import-dependent. No domestic production (net of re-exports).
- $0 < \mathrm{IDR} < 1$: Partial dependency. Higher values = greater import reliance.
- $\mathrm{IDR} > 1$: Country re-exports more than it produces domestically—it acts as a **trade hub**. This is itself a fraud-relevant signal (transit creates opacity).

Boundary condition: If $\mathrm{DS}'(c,i,t) \leq 0$, the IDR is undefined. This implies a data quality issue (stock drawdown or reporting error). Flag and exclude from scoring until resolved.

2.3 Origin Country Share (OCS)

Before defining bilateral dependency, we need the share that a specific origin contributes to total imports.

Definition 2.3 (Origin Country Share)**.** The proportion of country i's total imports of commodity c that is sourced from origin j:

$$\boxed{\mathrm{OCS}(c,i,j,t) \;=\; \frac{M(c,i,j,t)}{M(c,i,\bullet,t)} \;=\; \frac{M(c,i,j,t)}{\sum_{k\in\mathcal{O}} M(c,i,k,t)}} \tag{4}$$

By construction, $\sum_{j\in\mathcal{O}} \mathrm{OCS}(c,i,j,t) = 1$ and $\mathrm{OCS} \in [0,1]$.

2.4 Bilateral Dependency Index (BDI)

Definition 2.4 (Bilateral Dependency Index)**.** The share of country i's total domestic supply of commodity c sourced specifically from origin j:

$$\boxed{\mathrm{BDI}(c,i,j,t) \;=\; \frac{M(c,i,j,t)}{\mathrm{DS}'(c,i,t)}} \tag{5}$$

This decomposes cleanly into two distinct vulnerability drivers:

$$\mathrm{BDI}(c,i,j,t) \;=\; \underbrace{\mathrm{IDR}(c,i,t)}_{\substack{\text{overall import}\\ \text{dependency}}} \times \underbrace{\mathrm{OCS}(c,i,j,t)}_{\substack{\text{origin}\\ \text{concentration}}} \tag{6}$$

This decomposition is analytically critical. A country can have high IDR but low BDI for any single origin (diversified suppliers). Conversely, moderate IDR with very high OCS for one origin = concentrated dependency = the vulnerability.

Property 2.2 (BDI Interpretation)**.**

- $\mathrm{BDI} \approx 0$: Negligible dependency on this specific origin.
- $\mathrm{BDI} \approx 1$: Nearly the entire domestic supply comes from this single origin.
- $\mathrm{BDI} = \mathrm{IDR}$: This origin is the sole supplier. Maximum concentration risk.

2.5 Import Concentration: Herfindahl–Hirschman Index (HHI)

To measure how concentrated a country's import sources are across *all* origins:

Definition 2.5 (Import Concentration Index).

$$\boxed{\mathrm{HHI}(c,i,t) \;=\; \sum_{j\in\mathcal{O}} \Big[\mathrm{OCS}(c,i,j,t)\Big]^2 \;=\; \sum_{j\in\mathcal{O}} \left[\frac{M(c,i,j,t)}{M(c,i,\bullet,t)}\right]^2} \tag{7}$$

Property 2.3 (HHI Properties).

- $\mathrm{HHI} = 1/n$ for n equal suppliers: perfectly diversified.
- $\mathrm{HHI} = 1$: single-supplier monopoly.
- $\mathrm{HHI} > 0.25$: highly concentrated (standard antitrust threshold). A small number of origins dominate supply, reducing bargaining power and substitutability.

The HHI captures *systemic* vulnerability beyond what a single-corridor BDI shows. Even if no single origin dominates, high HHI means the market is oligopolistic—loss of any supplier has disproportionate impact.

2.6 Self-Sufficiency Ratio (SSR)

Definition 2.6 (Self-Sufficiency Ratio).

$$\boxed{\mathrm{SSR}(c,i,t) \;=\; \frac{P(c,i,t)}{D(c,i,t)}} \tag{8}$$

where $D(c,i,t)$ is FAOSTAT domestic supply for food use, or $\mathrm{DS}'(c,i,t)$ as proxy.

Property 2.4 (SSR Interpretation).

- $\mathrm{SSR} > 1$: Net exporter. Low structural vulnerability.
- $\mathrm{SSR} = 1$: Balanced but fragile.
- $\mathrm{SSR} < 1$: Gap of $(1 - \mathrm{SSR})$ must be filled by imports.
- $\mathrm{SSR} = 0$: No domestic production. Fully import-dependent.

2.7 Supply Criticality Index (SCI)

The SCI is the core corridor-specific measure that integrates import dependency, origin concentration, and market structure:

Definition 2.7 (Supply Criticality Index)**.** For corridor (c, i, j) at time t:

$$\boxed{\mathrm{SCI}(c,i,j,t) \;=\; \mathrm{IDR}(c,i,t) \;\times\; \mathrm{OCS}(c,i,j,t) \;\times\; \Big(1 + \mathrm{HHI}(c,i,t)\Big)} \tag{9}$$

The $(1+\mathrm{HHI})$ term serves as an **amplifier**: when the overall import market is concentrated, the vulnerability of any single corridor within that market is amplified because fewer alternative suppliers exist.

Property 2.5 (SCI Properties)**.**

- $\mathrm{SCI} \in [0,\, 2]$ since $\mathrm{IDR}, \mathrm{OCS} \in [0, 1]$ and $\mathrm{HHI} \in [0, 1]$, giving $\max = 1 \times 1 \times 2 = 2$.
- $\mathrm{SCI} = 0$: No dependency on this origin.
- $\mathrm{SCI} = 2$: Maximum vulnerability—fully import-dependent, sole supplier, zero diversification.
- For normalisation to $[0, 1]$: $\mathrm{SCI}_{\mathrm{norm}} = \mathrm{SCI}/2$.

Worked Example — RASFF 2026.0129: Cadmium in flaxseed, France → Belgium

Suppose (to be populated with actual Comtrade/FAOSTAT data):

Variable	Value	Source
$P(\mathrm{flax}, \mathrm{BE}, 2025)$	500 t	FAOSTAT
$M(\mathrm{flax}, \mathrm{BE}, \bullet, 2025)$	12,000 t	Comtrade
$X(\mathrm{flax}, \mathrm{BE}, \bullet, 2025)$	1,500 t	Comtrade
$M(\mathrm{flax}, \mathrm{BE}, \mathrm{FR}, 2025)$	8,000 t	Comtrade

Step 1: Apparent domestic supply

$$\mathrm{DS}' = 500 + 12{,}000 - 1{,}500 = 11{,}000 \text{ tonnes}$$

Step 2: Import Dependency Ratio

$$\mathrm{IDR} = \frac{12{,}000}{11{,}000} = 1.091 \quad (> 1\text{: Belgium is a transit hub for flaxseed})$$

Step 3: Origin Country Share

$$\mathrm{OCS} = \frac{8{,}000}{12{,}000} = 0.667$$

Step 4: HHI (suppose other origins: NL=2000t, DE=1500t, other=500t)

$$\mathrm{HHI} = (0.667)^2 + (0.167)^2 + (0.125)^2 + (0.042)^2 = 0.445 + 0.028 + 0.016 + 0.002 = 0.490$$

Step 5: Supply Criticality Index

$$\mathrm{SCI} = 1.091 \times 0.667 \times (1 + 0.490) = 1.091 \times 0.667 \times 1.490 = \mathbf{1.084}$$

$$\mathrm{SCI}_{\mathrm{norm}} = 1.084 \,/\, 2 = \mathbf{0.542}$$

Interpretation: Belgium has moderately high structural dependency on France for flaxseed. Its ability to reject or divert problematic French flaxseed shipments is constrained by supply needs.

3. Consumption Demand Modelling

The dependency models capture the supply side. The consumption models capture the demand side: how important this commodity is to the country's food system. Together they define the market pressure that fraud exploits.

3.1 Per Capita Apparent Consumption

$$\boxed{\mathrm{PCC}(c,i,t) \;=\; \frac{D(c,i,t)}{\mathrm{Pop}(i,t)}} \tag{10}$$

expressed in kg/capita/year. FAOSTAT Food Balance Sheets provide this directly for many commodity–country pairs.

3.2 Commodity Consumption Rank Score

For each country i and time t, rank all commodities by PCC:

Definition 3.1 (Consumption Rank Score)**.**

$$\boxed{\mathrm{CRS}(c,i,t) \;=\; 1 - \frac{\mathrm{Rank}(c,i,t) - 1}{|\mathcal{C}| - 1}} \tag{11}$$

where $\mathrm{Rank}(c,i,t)$ is the position of c when all commodities are ordered by $\mathrm{PCC}(c,i,t)$ descending, and $|\mathcal{C}|$ is the total number of commodities.

CRS = 1 for the highest-consumed commodity, CRS = 0 for the lowest. The normalised rank is preferable to raw PCC for cross-commodity comparison because different commodities operate on vastly different consumption scales.

3.3 Demand Inelasticity Indicator

Culturally embedded commodities (olive oil in Mediterranean countries, mussels in France) have inelastic demand—consumers buy them regardless of conditions. This makes the market exploitable. We estimate this using the coefficient of variation of historical consumption:

Definition 3.2 (Demand Inelasticity Score)**.**

$$\mathrm{CV}_D(c,i) \;=\; \frac{\sigma\big[\mathrm{PCC}(c,i,t)\big]}{\mu\big[\mathrm{PCC}(c,i,t)\big]} \qquad \text{over } t \in \{t{-}5,\,\ldots,\,t\} \tag{12}$$

$$\boxed{\mathrm{DIS}(c,i) \;=\; 1 - \min\Big(\mathrm{CV}_D(c,i),\;1\Big)} \tag{13}$$

DIS = 1 for perfectly stable demand (maximum exploitability). DIS = 0 for highly volatile demand.

4. Hazard Signal Modelling (RASFF–Trade Corridor Integration)

4.1 Corridor Definition from RASFF Data

A single RASFF notification r maps to one or more corridors. From the data structure:

$$\begin{aligned} \text{Origin}(r) &= j \quad \text{(the origin field)} \\ \text{Commodity}(r) &= c \quad \text{(mapped to HS code via commodity_code or NLP)} \\ \text{Affected}(r) &= \{i_1, i_2, \ldots, i_k\} \end{aligned}$$

where the Affected set is the union of:

$$\text{Affected}(r) = \underbrace{\text{notifying_country}}_{\text{detected the hazard}} \cup \underbrace{\text{for_attention}}_{\text{product may be on market}} \cup \underbrace{\text{for_followUp}}_{\text{investigation required}}$$

Each tuple (c, i_k, j) where $i_k \in \text{Affected}(r)$ defines a corridor touched by notification r.

Critical: A single RASFF notification generates multiple corridor signals. RASFF 2026.0053 (zearalenone in organic corn from France, 7 attention countries) creates 7 distinct corridors, each with its own dependency, consumption, and trade profile.

4.2 Severity Weighting

Not all notifications represent equal risk. The system assigns severity weights based on two dimensions from the RASFF data.

4.2.1 Classification Weight

Classification	W_{class}	Rationale
Alert notification	1.0	Product on market; immediate risk
Border rejection	0.8	Caught at border; indicates origin-level problem
Information for follow-up	0.7	Requires investigation
Information for attention	0.5	Informational; may not have reached market

4.2.2 Risk Decision Weight

Risk Decision	W_{risk}	Rationale
Serious	1.0	Confirmed serious health risk
Potentially serious	0.7	Risk under certain conditions
Potential risk	0.4	Risk not confirmed but not excluded
Not serious	0.2	Minor or no health impact

Definition 4.1 (Notification Severity Weight).

$$\boxed{S(r) \;=\; W_{\text{class}}(r) \;\times\; W_{\text{risk}}(r)} \tag{14}$$

yielding $S(r) \in [0.1,\ 1.0]$.

4.3 Hazard Intensity Score (HIS)

The HIS aggregates all RASFF notifications for a corridor, weighted by severity and temporal recency:

Definition 4.2 (Hazard Intensity Score).

$$\boxed{\mathrm{HIS}(c,i,j,t) \;=\; \sum_{r\,\in\,\mathcal{R}(c,i,j)} S(r) \;\cdot\; \alpha^{\,(t-t_r)}} \tag{15}$$

where:

$$\begin{aligned} \mathcal{R}(c,i,j) &= \text{set of all RASFF notifications touching corridor } (c,i,j) \\ \alpha \in (0,1) &= \text{temporal decay parameter} \\ t_r &= \text{time period of notification } r \end{aligned}$$

The decay parameter α controls the half-life of a notification's influence:

$$t_{1/2} = \frac{-\ln 2}{\ln \alpha} \tag{16}$$

α	**Half-life (monthly periods)**	**Use case**
0.95	$\approx$ 13.5 months	Shelf-stable commodities
0.90	$\approx$ 6.6 months	General-purpose default
0.85	$\approx$ 4.3 months	Perishable commodities

The HIS is **intentionally unbounded above** at this stage. Corridors with many high-severity notifications *should* produce large scores. Normalisation happens at the composite scoring stage (§7).

4.4 Hazard Diversity Index (HDI)

A corridor with multiple different hazard types suggests systemic quality failures, not a single controllable problem. Using Shannon entropy:

Definition 4.3 (Hazard Diversity Index).

$$\boxed{\mathrm{HDI}(c,i,j) \;=\; -\sum_{h\in\mathcal{H}} p_h \cdot \ln p_h} \tag{17}$$

where p_h is the proportion of notifications for corridor (c,i,j) with hazard type h. Normalised to $[0,1]$:

$$\mathrm{HDI}_{\mathrm{norm}}(c,i,j) \;=\; \frac{\mathrm{HDI}(c,i,j)}{\ln|\mathcal{H}|} \tag{18}$$

$\mathrm{HDI}_{\mathrm{norm}} = 0$: all notifications have the same hazard type (no diversity).
$\mathrm{HDI}_{\mathrm{norm}} = 1$: all hazard types equally represented (maximum diversity).

4.5 Detection Gap Indicator (DGI)

A corridor with high trade volume but zero RASFF notifications may be under-inspected, not safe.

Definition 4.4 (Detection Gap Indicator).

$$\boxed{\mathrm{DGI}(c,i,j,t) \;=\; \frac{M(c,i,j,t)}{M(c,i,\bullet,t)} \;-\; \frac{R(c,i,j,t)}{R(c,i,\bullet,t)}} \tag{19}$$

Property 4.1 (DGI Interpretation).

- DGI > 0: This origin supplies a larger share of trade than it accounts for in notifications. Either genuinely cleaner, or **under-inspected**.
- DGI < 0: Over-represented in problems relative to trade share.
- DGI > 0 combined with high BDI: **flag for potential under-detection**.

DGI is undefined when $R(c,i,\bullet,t) = 0$. In this case, flag as "insufficient data."

5. Trade Flow Analysis Models

These models operate on UN Comtrade data to detect anomalies in trade patterns that may indicate fraudulent activity.

5.1 Unit Value Analysis (Price Proxy)

Definition 5.1 (Unit Value and Anomaly Score).

$$\mathrm{UV}(c,i,j,t) \;=\; \frac{V(c,i,j,t)}{M(c,i,j,t)} \quad (\mathrm{USD/kg}) \tag{20}$$

Cross-origin mean and standard deviation:

$$\mu_{\mathrm{UV}}(c,i,t) = \frac{1}{|\mathcal{O}_{\mathrm{active}}|}\sum_{j}\mathrm{UV}(c,i,j,t) \tag{21}$$

$$\sigma_{\mathrm{UV}}(c,i,t) = \sqrt{\frac{1}{|\mathcal{O}_{\mathrm{active}}|-1}\sum_{j}\Big[\mathrm{UV}(c,i,j,t)-\mu_{\mathrm{UV}}(c,i,t)\Big]^2} \tag{22}$$

Z-score for origin j:

$$\boxed{Z_{\mathrm{UV}}(c,i,j,t) \;=\; \frac{\mathrm{UV}(c,i,j,t)-\mu_{\mathrm{UV}}(c,i,t)}{\sigma_{\mathrm{UV}}(c,i,t)}} \tag{23}$$

Property 5.1 (Price Anomaly Signals).

- $Z_{\mathrm{UV}} < -2$: price well below market—potential adulteration, substitution, or mis-declaration.
- $Z_{\mathrm{UV}} > +2$: price well above market—potential premium fraud (false quality/origin claims).

5.2 Volume Anomaly Detection

Rolling-window approach using the corridor's own history:

Definition 5.2 (Volume Anomaly Score). Over a k-period rolling window:

$$\mu_M(c,i,j) = \frac{1}{k}\sum_{\tau=t-k}^{t-1} M(c,i,j,\tau) \tag{24}$$

$$\sigma_M(c,i,j) = \sqrt{\frac{1}{k-1}\sum_{\tau=t-k}^{t-1}\Big[M(c,i,j,\tau)-\mu_M(c,i,j)\Big]^2} \tag{25}$$

$$\boxed{Z_M(c,i,j,t) \;=\; \frac{M(c,i,j,t)-\mu_M(c,i,j)}{\sigma_M(c,i,j)}} \tag{26}$$

$Z_M > 2$ indicates a trade surge warranting investigation for re-routing, fraudulent volume inflation, or origin laundering.

5.3 Mirror Trade Discrepancy

Trade data is reported by both importer and exporter. Discrepancies may indicate fraud.

Definition 5.3 (Mirror Trade Discrepancy).

$$\boxed{\mathrm{MTD}(c,i,j,t) \;=\; \frac{\big|M_i(c,i,j,t) \;-\; X_j(c,j,i,t)\big|}{\max\Big(M_i(c,i,j,t),\;\; X_j(c,j,i,t)\Big)}} \tag{27}$$

where M_i is what country i reports importing from j, and X_j is what country j reports exporting to i.

Typical legitimate discrepancy (CIF/FOB valuation, timing): 5–15%. Discrepancies exceeding 30% warrant investigation. Persistent large discrepancies over multiple periods = strong fraud signal.

5.4 Trade Network Concentration Shift

Monitor structural changes in import concentration over time:

$$\boxed{\Delta\mathrm{HHI}(c,i,t) \;=\; \mathrm{HHI}(c,i,t) \;-\; \mathrm{HHI}(c,i,t{-}1)} \tag{28}$$

$$\boxed{\Delta\mathrm{OCS}(c,i,j,t) \;=\; \mathrm{OCS}(c,i,j,t) \;-\; \mathrm{OCS}(c,i,j,t{-}1)} \tag{29}$$

A sudden increase in HHI (market becoming more concentrated) combined with a new or rapidly growing corridor = potential re-routing signal. A rapid increase in OCS for a previously minor origin, while a traditionally dominant origin's OCS decreases, is a structural shift warranting investigation.

6. Origin–Attention Country Relationship Model

This section formalises the network structure of hazard exposure propagation across the EU's trade system.

6.1 Exposure Network as a Directed Graph

Definition 6.1 (Exposure Network)**.** Define a directed weighted graph $G = (V, E)$ where:

$$V = \mathcal{O} \cup \mathcal{N} \quad \text{(all origin and destination countries)}$$
$$E \subseteq \mathcal{O} \times \mathcal{N} \quad \text{(directed edges from origin to destination)}$$

An edge $(j \to i)$ exists if origin j has triggered a RASFF notification where country i appeared as notifier, attention, or follow-up. Each edge carries commodity-specific weights:

$$w_{\text{trade}}(j, i, c, t) = M(c, i, j, t) \tag{30}$$
$$w_{\text{hazard}}(j, i, c, t) = \text{HIS}(c, i, j, t) \tag{31}$$
$$w_{\text{dep}}(j, i, c, t) = \text{BDI}(c, i, j, t) \tag{32}$$

6.2 Origin Risk Propagation Score (ORPS)

For an origin country j: how broadly does its food fraud risk propagate across the EU?

Definition 6.2 (Origin Risk Propagation Score)**.**

$$\boxed{\text{ORPS}(j, c, t) \;=\; \sum_{i \in \mathcal{N}} \text{BDI}(c, i, j, t) \;\cdot\; \text{HIS}(c, i, j, t) \;\cdot\; \text{PCC}(c, i, t)} \tag{33}$$

This sums, across all EU destination countries, the product of bilateral dependency, hazard intensity, and per capita consumption. The result captures **total population-weighted vulnerability exposure** created by this origin for this commodity across the entire EU. Origins with high ORPS should receive coordinated EU-level action.

6.3 Attention Country Exposure Profile (ACEP)

For an attention country i: what is its total inbound fraud exposure?

Definition 6.3 (Attention Country Exposure Profile)**.**

$$\boxed{\text{ACEP}(i, t) \;=\; \sum_{c \in \mathcal{C}} \sum_{j \in \mathcal{O}} \text{BDI}(c, i, j, t) \;\cdot\; \text{HIS}(c, i, j, t) \;\cdot\; \text{CRS}(c, i, t)} \tag{34}$$

Countries with high ACEP need more inspection resources, laboratory capacity, and coordination support.

6.4 Conditional Probability of Hazard Given Trade

For corridors with sufficient RASFF history (≥ 10 notifications over 5 years):

Definition 6.4 (Empirical Hazard Probability).

$$\boxed{\hat{P}(\text{hazard} \mid c, i, j) \;=\; \frac{R(c, i, j, \mathcal{T})}{\dfrac{M(c, i, j, \mathcal{T})}{\overline{m}(c)}}} \tag{35}$$

where $\mathcal{T}$ is the total observation period, $R(c, i, j, \mathcal{T})$ is the total notification count, and $\overline{m}(c)$ is the estimated average shipment size for commodity c.

This is inherently a **lower bound**—it only captures detected hazards. Cross-reference with the Detection Gap Indicator (§19) to distinguish "more fraud" from "more detection."

7. Composite Vulnerability Scoring

7.1 Normalisation

Each model output operates on a different scale. Before combination, they must be normalised. Three defensible approaches:

7.1.1 Option A: Min-Max Normalisation

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{36}$$

Maps to $[0, 1]$. Simple but sensitive to outliers. $x_{\min}$ and $x_{\max}$ are computed over all corridors.

7.1.2 Option B: Percentile Rank Normalisation

$$x_{\text{norm}} = \text{percentile_rank}(x) \quad \text{across all corridors} \tag{37}$$

Maps to $[0, 1]$. Robust to outliers. Interpretation: "this corridor is in the 85th percentile for this metric." More stable when underlying distributions are skewed (which they will be—most corridors are low-risk).

7.1.3 Option C: Log-Transform then Percentile

$$x_{\text{norm}} = \text{percentile_rank}\Big(\ln(1 + x)\Big) \tag{38}$$

Appropriate for highly skewed metrics (notification counts, trade volumes) where a small number of corridors dominate.

Recommendation: Option B (percentile rank) for most metrics. Option C (log-percentile) for HIS due to exponential distribution of notification counts. Validate with your data once a full corridor dataset is available.

7.2 Composition Approaches

Three approaches, each with different properties:

7.2.1 Approach 1: Weighted Linear Combination (Compensatory)

$$\text{CVS} = \sum_{k=1}^{K} w_k \cdot x_{k,\text{norm}} \qquad \text{where} \sum_{k=1}^{K} w_k = 1 \tag{39}$$

A high score on one metric compensates for a low score on another. A corridor with zero hazard history but extreme dependency would still get a moderate score. Whether this is desirable is a design decision.

7.2.2 Approach 2: Geometric Mean (Non-Compensatory)

$$\text{CVS} = \prod_{k=1}^{K} \left(x_{k,\text{norm}}\right)^{w_k} \tag{40}$$

If *any* component is zero, the composite is zero. Appropriate if you believe all conditions must be present for vulnerability. More conservative; may miss emerging risks.

7.2.3 Approach 3: Hybrid—Structural Base with Signal Amplifier

$$\boxed{\text{CVS} = \underbrace{\text{SCI}_{\text{norm}} \times \text{CRS}_{\text{norm}}}_{\text{BASE: structural vulnerability}} \times \left(1 + \underbrace{w_h \cdot \text{HIS}_{\text{norm}} + w_p \cdot \text{PAS}_{\text{norm}} + w_{sc} \cdot \text{SCCS}_{\text{norm}}}_{\text{AMPLIFIER: hazard \& trade signals}}\right)} \tag{41}$$

The BASE requires both structural dependency and consumption demand. If either is zero, the corridor scores zero regardless of other factors—you cannot have vulnerability without a market to exploit. The AMPLIFIER adds risk weight from hazard history, price anomalies, and supply chain complexity, but cannot *create* vulnerability where structural conditions do not exist.

$$\text{CVS}_{\text{hybrid}} \in \Big[0,\ \ 1 \cdot (1 + w_h + w_p + w_{sc})\Big]$$

Normalise to $[0, 1]$ by dividing by the theoretical maximum $(1 + w_h + w_p + w_{sc})$.

All three approaches are presented because the right choice depends on a decision you must make: should the system flag corridors where structural vulnerability exists but no hazard has been detected yet (Approach 3), or only corridors where all signals converge (Approach 2)?

7.3 Weight Determination

7.3.1 Option A: Expert Elicitation (Analytic Hierarchy Process)

Convene food fraud investigators and domain experts. Use structured pairwise comparison (AHP) to derive weights from expert judgement. Most defensible starting point without validated outcome data.

7.3.2 Option B: Equal Weights

$$w_k = \frac{1}{K} \quad \forall\, k \tag{42}$$

Most transparent and least assumption-laden. Research in multi-criteria decision analysis shows equal-weight models often perform surprisingly close to optimally-weighted models when true optimal weights are unknown.

7.3.3 Option C: Empirical Calibration (Future)

Once sufficient validated fraud outcomes are accumulated ($n \geq 50$–100 across diverse commodities):

$$\hat{P}(\text{fraud} \mid \mathbf{x}) = \frac{1}{1 + \exp\big(-(\beta_0 + \beta_1 x_1 + \cdots + \beta_K x_K)\big)} \tag{43}$$

The learned logistic regression coefficients $\hat{\beta}_k$ serve as empirically calibrated weights. Alternative: gradient-boosted classification (XGBoost/LightGBM) with SHAP values for interpretability.

Honest assessment: Option C will not be available at launch. Start with Option A or B. Equal weights with expert review of the first batch of scored corridors is the pragmatic path.

8. What This Framework Does Not Yet Cover

The following are areas where the mathematical modelling requires input that has not been specified:

1. **Price data source and granularity.** The unit value analysis (§5.1) uses Comtrade value/volume as a price proxy. This is crude. If actual import price indices (Eurostat) or wholesale market prices are available, the models become significantly more precise. What price data do you have or plan to acquire?

2. **HS code mapping methodology.** The entire framework depends on reliably matching RASFF commodity descriptions to HS codes for Comtrade matching. The `Commodity_Code` field in your data is sometimes populated, sometimes not. What is the plan for missing values? This is a make-or-break data engineering problem.

3. **Temporal granularity.** The models work at annual or monthly resolution. RASFF data is daily; Comtrade is annual/monthly; FAOSTAT is annual with 1–2 year lag. What is the target update cycle?

4. **Validation data.** Do you have access to confirmed fraud cases (from FFN, OLAF, or Member State authorities) to validate model outputs? Without validation, the scoring system is a hypothesis, not a proven model.

5. **Scope of commodities.** Your sample data covers cereals, feed materials, and bivalve molluscs. Different commodity types may require different parameterisations (e.g., the decay parameter α should be faster for perishable goods than for shelf-stable goods).

END OF MATHEMATICAL FRAMEWORK | Version 1.0 | February 2026